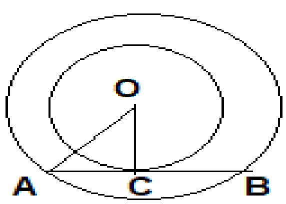
PRE- BOARD EXAMINATION (2024-25)

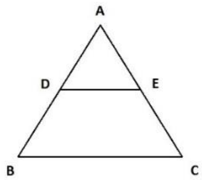
CLASS – X

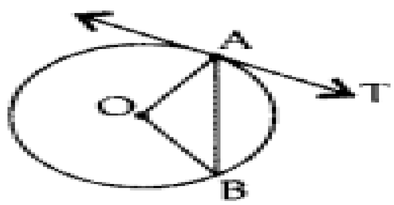
TIME: 3 Hrs SUBJECT: MATHEMATICS STANDARD (041) M.M: 80

INSTRUCTIONS:

This paper is divided into 5 sections. Section A contains 18 multiple choice questions and 2 assertion reason type questions of 1 mark each and all questions are compulsory.
Section B contains 5 very short answer type questions of 2 marks each.
Section C contains 6 short answer type questions of 3 marks each.
Section D contains 4 long answer type questions of 5 marks each.
Section E contains 3 CCT type questions of 4 marks each.
There is an internal choice in 2 questions of section B, 2 questions of section C and 2 questions of section D. Student has to attempt only one in such questions.

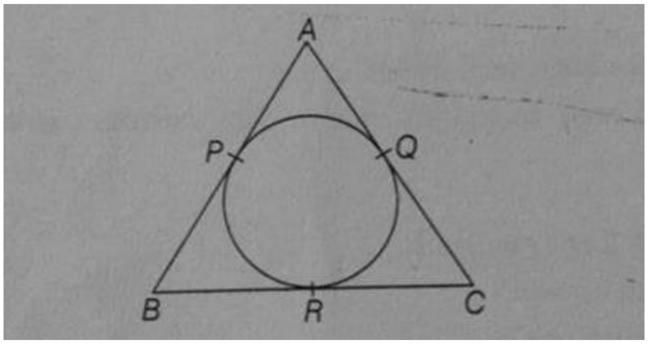
S.No.	QUESTION	MARKS
	SECTION A [1X20 =20]	
1	In the fig. two concentric circles of radii a and b ($a > b$) are given. The chord AB of larger circle touches the smaller circle at C. Find the length of AB.  (a) $2\sqrt{b^2 - a^2}$ (b) $2\sqrt{a^2 - b^2}$ (c) $2\sqrt{a^2 - 2b^2}$ (d) $2\sqrt{2a^2 - b^2}$	1
2	If y =1 is one of the solutions of the quadratic equation $py^2 + py + 3 = 0$, then the value of p is: (a) -3 (b) 2 (c) -3/2 (d) -2	1
3	If a polynomial p(x) is given by $p(x) = x^2 – 5x + 6$, then the value of p(1) + p(4) is: (a) 0 (b) 4 (c) 2 (d) -4	1
4	The ratio in which the line segment joining the points A(-2, -3) and B(3, 7) is intersected internally by the y axis is: (a) 3:2 (b) 2:3 (c) 3:7 (d) 7:3	1
5	The fourth vertex of a parallelogram ABCD whose three vertices are A(-2,3), B(6,7), and C(8,3) is: (a) (0,1) (b) (0,-1) (c) (-1,0) (d) (1,0)	1
6	The point on x axis which is equidistant from the points (5,-3)and (4,2) is: a) (4.5,0) b) (7,0) c) (0.5,0) d) (-7,0)	1
7	The value of the expression $\sin^2\theta + 1/(1+ \tan^2 \theta)$ is : (a) 0 (b) 2 (c) 1 (d) -1	1
8	The sum of first 20 odd natural numbers is: (a) 150 (b) 200 (c) 400 (d) 210	1

9	In ΔABC, DE ∥ BC as shown in the figure. If AD = 4cm, AB = 9 cm and AC = 13.5 cm, then  the length of EC is: (a) 6cm (b) 7.5 cm (c) 9 cm (d) 5.7 cm	1
10	If $3825 = 3^x \times 5^y \times 17^z$, then the value of x + y - 2z is: (a)40 (b) 28 (c) 25 (d)36	1
11	A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at point Q so that OQ = 12 cm. Length PQ is : (a) 12 cm (b) 13 cm (c) 8.5 cm (d) $\sqrt{119}$ cm	1
12	For the event E, if $P(E) + P(\bar{E}) = q$, then value of $q^2 - 4$ is: (a) -3 (b) 3 (c) 5 (d) -5	1
13	Two dice are tossed, the probability of getting doublets of odd numbers on both dice is (a) 3/12 (b) 1/12 (c) 5/12 (d) 7/12	1
14	The area of the sector of a circle of radius 12 cm is 60π cm². The central angle of this sector is: (a) 120^0 b) 60^0 c) 75^0 d) 150^0	1
15	If the length of an arc of a circle of diameter 84 cm is 88 cm, then the angle subtended by the arc at the centre of the circle is: (a) 120^0 (b) 90^0 (c) 60^0 (d) 30^0	1
16	At some time of the day, the length of the shadow of a tower is equal to its height. Then the Sun's altitude at that time is: 30^0 (b) 45^0 (c) 60^0 (d) 90^0	1
17	If two dice are thrown together, what is the probability of getting the sum of two numbers appearing on them as 7? (a)1/6 (b)3/5 (c)3/4 (d)1/2	1
18	If cos X = a/b, then sin X is equal to : (a)$(b^2-a^2)/b$ (b) (b-a)/b (c) $\sqrt{(b^2-a^2)}/b$ (d) $\sqrt{(b-a)}/b$	1
19.	**DIRECTIONS:** In the question number 19 and 20, a statement of A**ssertion (A)** is followed by a statement of **Reason (R)**. Choose the correct option Assertion (A) In an AP, $S_n = n^2+n$, then $a_{20} = 40$. Reason (R) In an AP, $a_n - a_{n-1} = d$ a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A) (b) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A) (c) Assertion (A) is true but reason(R) is false. (d) Assertion (A) is false but reason(R) is true	1
20.	Statement -1 (Assertion) : the curved surface area of a cone of base radius 6 cm and slant height 10 cm is is 60π cm^2 .	1

	Statement -2 (Reason) : curved susrface area of a cone = $\pi r^2 h$, where r is the radius and h is the height of the cone. a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A) (b) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A) (c) Assertion (A) is true but reason(R) is false. (d) Assertion (A) is false but reason(R) is true	
	SECTION B [2 X 5 = 10]	
21.	Prove that of $2-3\sqrt{5}$ is irrational .	2
22	If a line intersects sides PQ and PR of a triangle PQR at S and T respectively and ST is parallel to QR, prove that PQ/PS = PR/PT.	2
23	In the given figure, O is the centre of a circle, AB is a chord and AT is the tangent at A. If $\angle AOB = 100°$, then calculate $\angle BAT$.  OR Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that $\angle PTQ = 2\angle OPQ$	2
24	If the area of a sector of circle having radius 35 cm is 3850/4 cm^2, find the angle subtended by the corresponding arc at the centre of the circle. OR If the area swept by the minute hand of a clock in 5 minutes is 154/3 cm^2, find the length of the minute hand.	2
25	The shadow of a vertical tower is 60 m long at some instant. If the length of the tower is $20\sqrt{3}$ m, find the Sun's altitude at that instant.	2
	SECTION C [3 X 6 =18]	
26	Show that $(\sqrt{5} - \sqrt{3})$ is an irrational number.	3
27	If α and β are the zeroes of the polynomial $6x^2 - 7x + 2$, find the value of $\alpha^2 + \beta^2$	3
28	Solve for x and y: $a^2x + b^2y = c^2$ $b^2x + a^2y = d^2$ OR Solve the system of equations to get the value of x and y: $434x + 262y = 1826$ $262x + 434y = 1654$	3

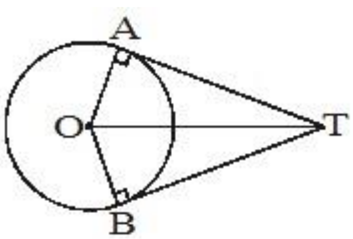
<table>
<tr><td>29</td><td>Prove that:
$\frac{\sin\theta - \cos\theta+1}{\sin\theta+\cos\theta-1} = \frac{1}{\sec\theta-\tan\theta}$</td><td>3</td></tr>
<tr><td>30</td><td>

In the figure if AP = PB, then show that AC = BC

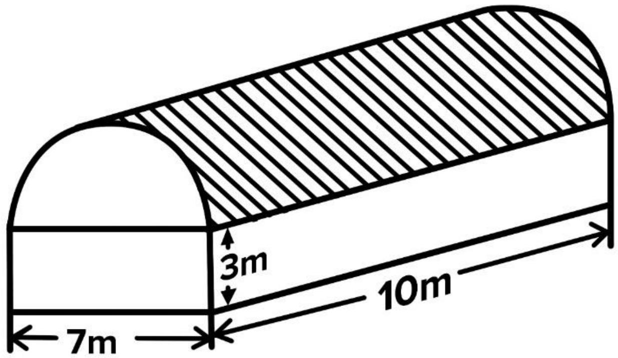


Or

In figure, If ∠ATO = 40°, find ∠AOB



</td><td>3</td></tr>
<tr><td>31</td><td>The median of the following frequency distribution is 14.4. Find the values of x and y, if the sum of frequencies is 20..
<table>
<tr><td>Class-interval</td><td>0-6</td><td>6-12</td><td>12-18</td><td>18-24</td><td>24-30</td><td>Total</td></tr>
<tr><td>Frequency</td><td>4</td><td>x</td><td>5</td><td>y</td><td>1</td><td>20</td></tr>
</table>
</td><td>3</td></tr>
<tr><td colspan="3">SECTION D [5 X 4 = 20]</td></tr>
<tr><td>32</td><td>Find the value of k for which the quadratic equation $(k+1)x^2$ -2(3k + 1)x + (8k +1= 0 has real and equal roots.</td><td>5</td></tr>
<tr><td>33</td><td>State and prove Basic Proportionality Theorem.
By using the theorem if in a triangle KMN PQ is parallel to MN and If KP/PM= 4/13 , KN = 20.4 cm ,then find KQ.
OR
A girl of height 90cm is walking away from the base of a lamp post at a speed of 1.2 m/s. If the lamp is 3.6m above the ground, find the length of her shadow after 4 seconds.</td><td>5</td></tr>
<tr><td>34</td><td>A godown building is in the form as shown in the figure.</td><td>5</td></tr>
</table>



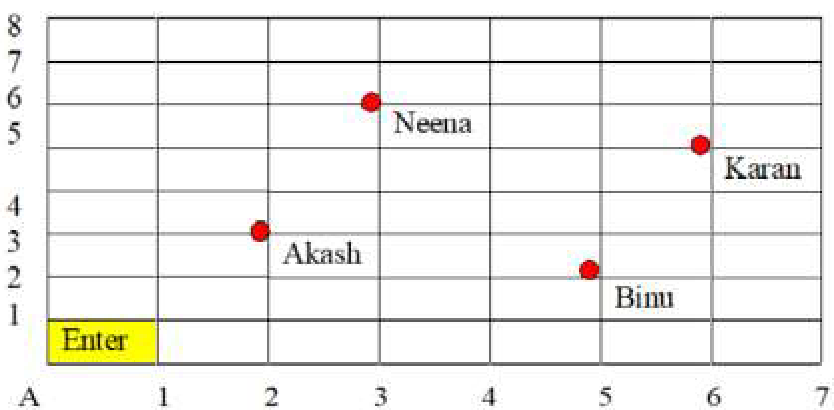
The vertical cross section parallel to the width of the building is a rectangle of dimensions 7m x 3mmounted by a semi circle of radius 3.5 m. the inner measurements of the cuboidal portion of the building are 10m x 7m x 3m. find the interior surface area excluding the floor.

35 The following distribution shows the daily pocket allowance of children of a locality. The mean daily pocket allowance is Rs. 36.10. find the missing frequency. 5

Daily pocket allowance	20-25	25-30	30-35	35-40	40-45	**45-50**	**50-55**
No.of children	7	6	9	13	F	**5**	**4**

SECTION E [4 X 3 =12] CASE STUDY BASED QUESTIONS

36. Karan went to the Lab near to his home for COVID 19 test along with his family members. The seats in the waiting area were as per the norms of distancing during this pandemic (as shown in the figure below). His family members took their seats surrounded by a red circular area 1+1+2



i) What is the distance between Neena and Karan?

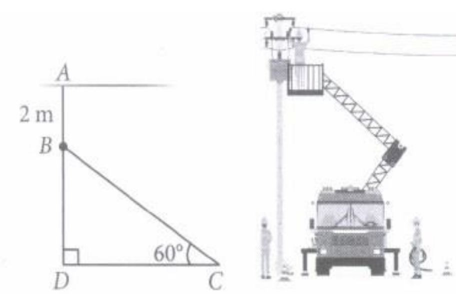
ii) What are the coordinates of the seat of Akash?

iii) What will be the coordinates of a point exactly between Akash and Binu where a person can be seated?

OR

Find the coordinates of mid point of Neenu and Binu

37. Nikhil started saving money for his new project in a BANK 1+1+2

	 . He started saving Rs 240 in the first month, Rs 300 in the second month Rs 360 in the third month and so on. He continues to save in this manner for quite some time. Based on the above, answer the following questions : (i) Are the numbers representing his savings in AP ? If so, write the first term (a) and the common difference(d). (ii) In which month will he save Rs 660 ? (iii)What amount will he save in the 15th month ? OR How much money he will accumulate after 10 months ?	
38	Electricians are responsible for inspecting, testing, repairing, installing, and modifying electrical components and systems. Electricians general work at homes, businesses, and construction sites, and generally work as contractors. An electrician has to repair an electric fault on the pole of height of 8 m. He needs to reach a point 2 m below the top of the pole to undertake the repair work.  Based on the above information, answer the following questions. i) Length of BD is ? ii) What should be the length of ladder, so that it makes an angle of 60° with the ground? **OR** The distance between the foot of ladder and pole is? iii) What will be the measure of ∠BCD when BD and CD are equal?	1+2+1